DATA SCIENCE CAPSTONE SHOWCASE

Predicting Contraceptive Use Among Kenyan Women

A Machine Learning & Deep Learning Approach to Targeting Family Planning Interventions

Prepared by: The Insight Architects

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Exploratory Data Analysis

Understanding the demographic, socioeconomic, and geographic landscape

Project Overview & Target Distribution

The Core Problem

Kenya's mCPR is ~60% nationally, hiding severe regional/socioeconomic disparities.

Goal: predict modern contraceptive use to help health planners target resources.

Target Variable (4-Class)

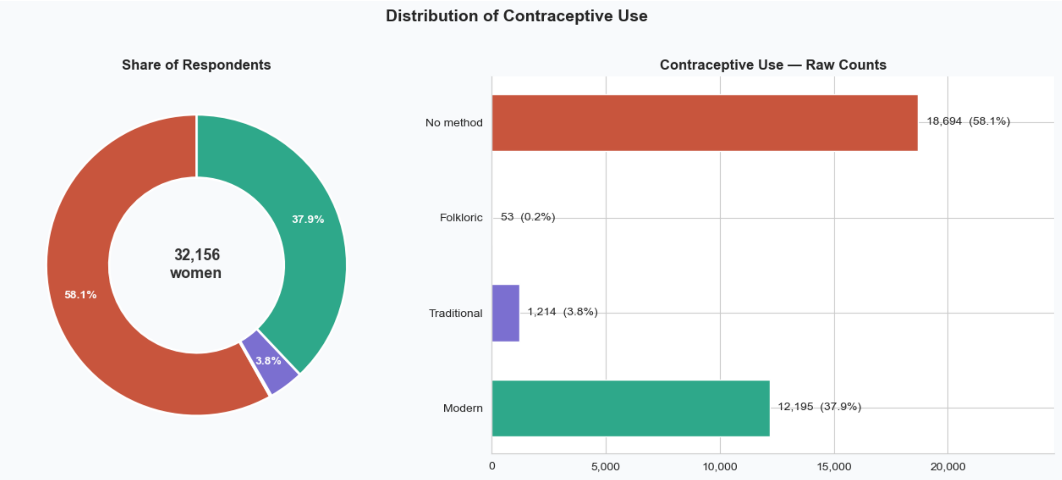
No method: 58.1% (18,694 women)

Modern method: 37.9% (12,195)

Traditional: 3.8% (1,214)

Folkloric: 0.2% (53 women)

The extreme class imbalance (Folkloric has only 53 cases) is a key challenge.



Demographics: Age Distribution

Age Profile

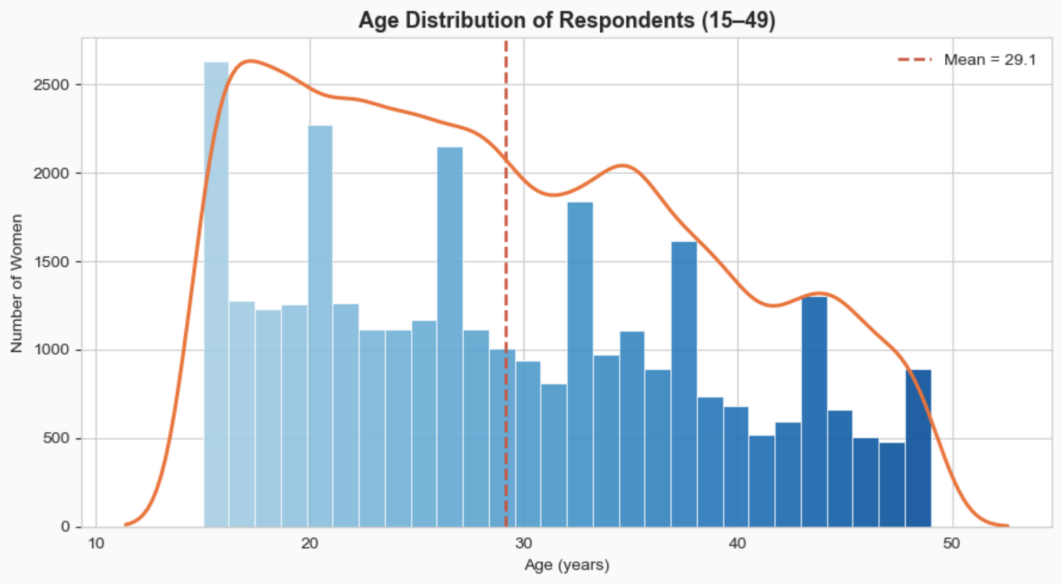
Mean age ≈ 29.1, median 28. Numbers thin toward the 49 upper bound.

KDHS 2022 sampled women aged 15–49 across all 47 counties.

Numeric Summary

Household size and living children are right-skewed.

```
print('Rows and columns:', df.shape)
df.describe().T
```



Demographics: Education Level

Education Level Distribution

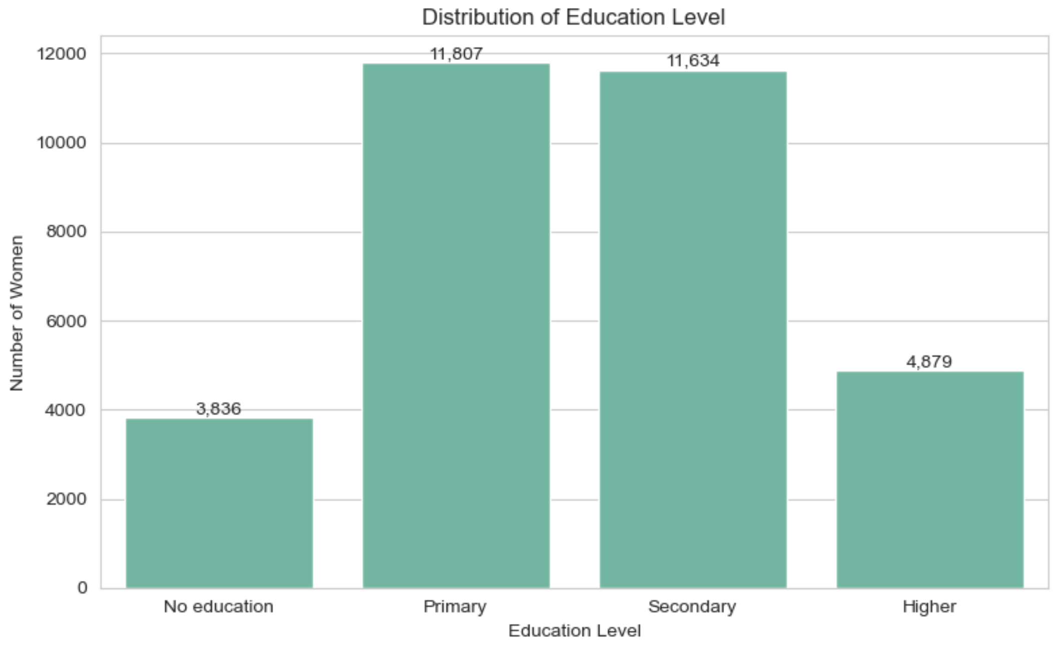
Primary: 36.7%

Secondary: 36.2%

Higher: 15.2%

No education: 11.9%

Education influences health awareness and access to reproductive health information.



Socioeconomics: Wealth Index & Residence

Wealth Index

Quintiles fairly balanced (18–22% each): Richer 22.3%, Poorest 22.0%, Middle 19.7%, Richest 18.1%, Poorer 17.9%.

Residence Type

Rural: 61.5%

Urban: 38.5%

Urban women have marginally higher resource access; rural majority is the policy focus.

Bivariate: Education as a Driver of Uptake

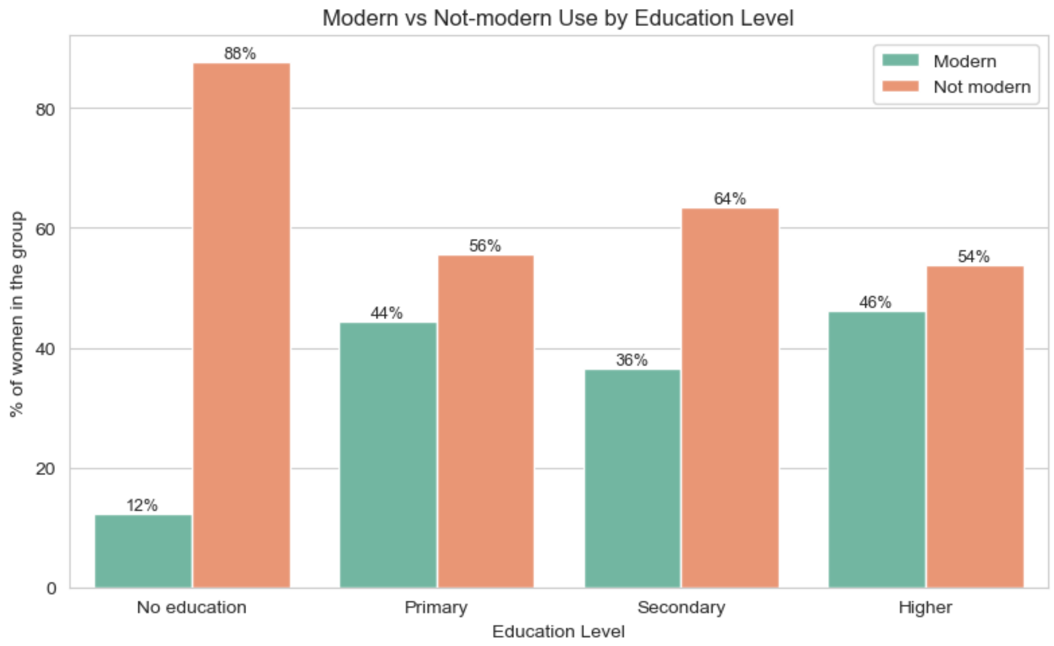
Education as a Catalyst

No education: ~12% modern use vs. any education: ~37–46%.

Primary-educated women (44%) are slightly higher than secondary (37%), likely due to age/parity confounding.

Key Finding

Even primary education significantly boosts modern method adoption.



Bivariate: Wealth as a Driver of Uptake

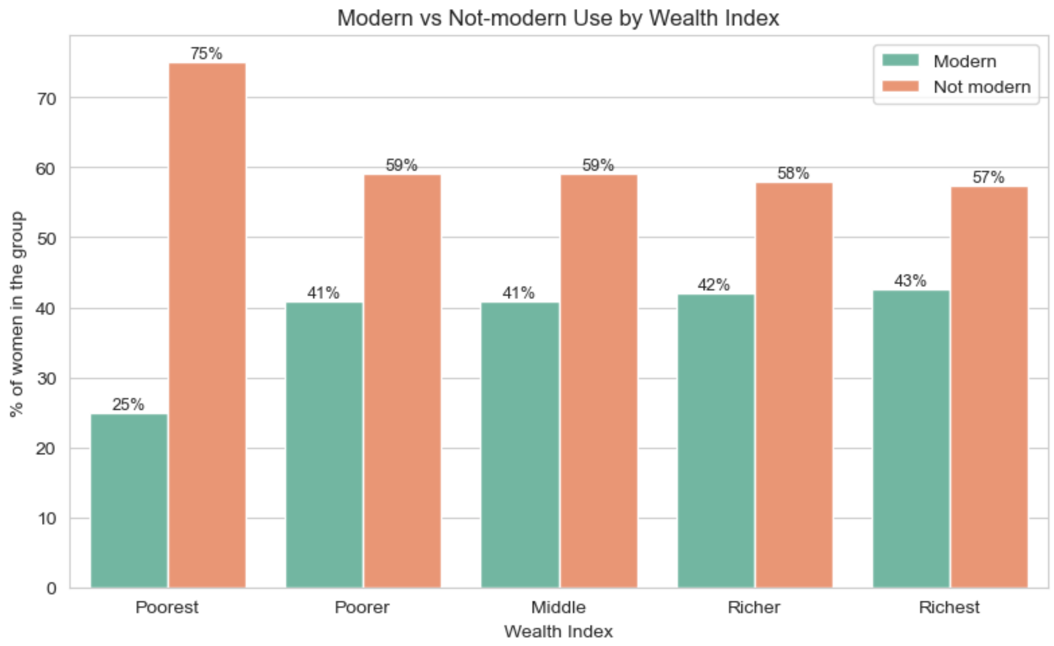
Wealth: A Cliff, Not a Gradient

Poorest quintile: ~25% modern use. Above that, flat at ~41–43%.

The big gap is between poorest and everyone else, not a steady climb from poor to rich.

Implication

Subsidies must target the Poorest quintile exclusively for maximum ROI.



Bivariate: Residence Type

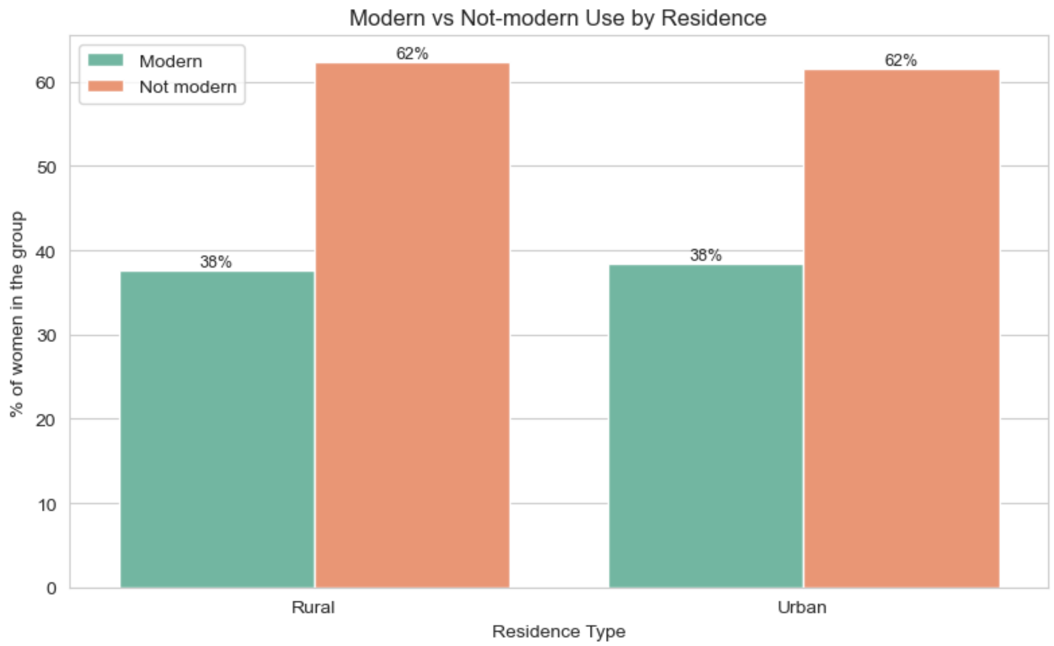
Urban vs. Rural

Modern use is almost identical: ~38% for both urban and rural women.

Key Finding

On its own, residence barely matters. Any urban/rural gap is explained by education and wealth, not geography alone.

This means rural interventions should focus on education, not just proximity.



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Geographic Disparities: Modern Use by County

Massive County Variation

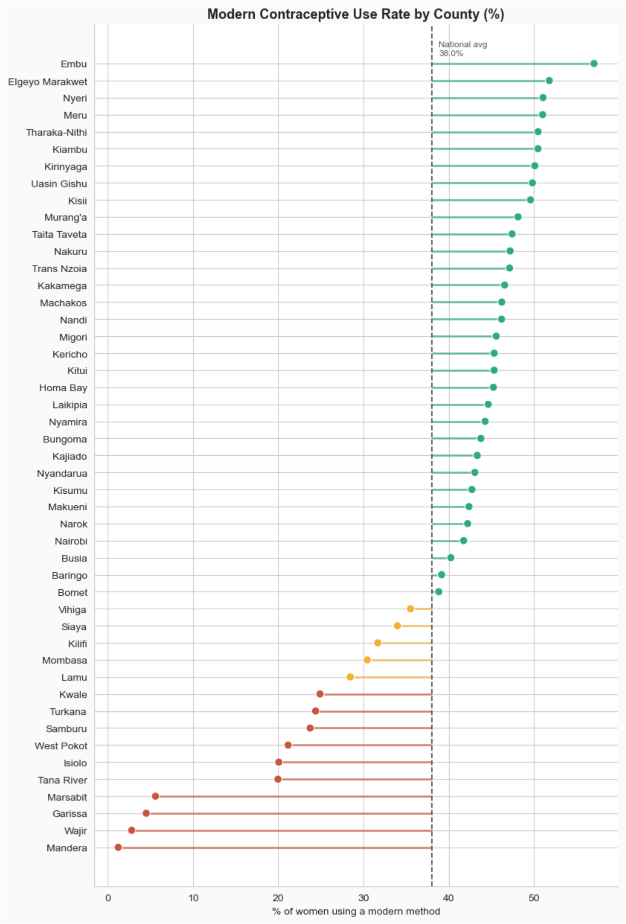
Lowest: Mandera (1.2%), Wajir (2.8%), Garissa (4.5%), Marsabit (5.6%)

Highest: Embu, Meru, Nyeri, Elgeyo Marakwet (50–57%)

All low-use counties are in the arid/semi-arid north (ASAL).

Implication

Geographically targeted interventions are essential — national averages mislead.



Multivariate: Education vs. Wealth Interaction

Education Trumps Wealth

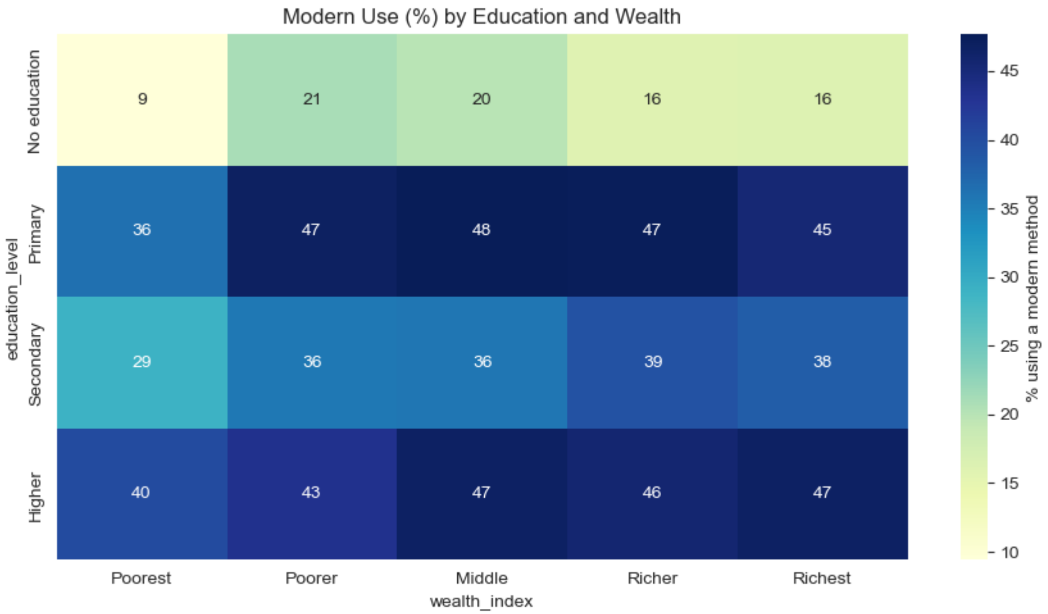
No education + any wealth: 9–21% modern use.

Any education + poorest: still ~35–48%.

Key Takeaway

Literacy and health education are far more critical than income for long-term family planning adoption.

Wealth cannot compensate for lack of education.



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Feature Correlation & Collinearity Audit

Correlation Matrix

children_ever_born and living_children: r = 0.98 (near-perfect collinearity).

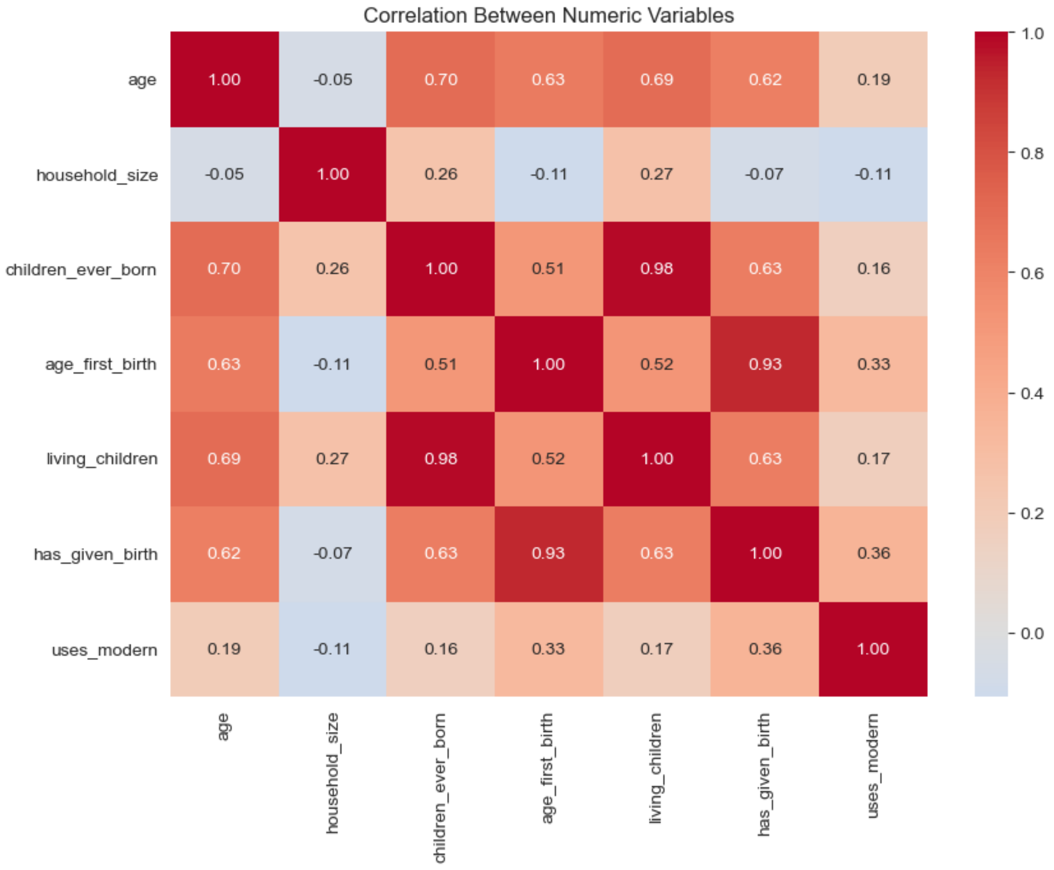
Resolved by deriving child_loss and surviving_ratio.

Other Observations

Age ↔ children: ~0.65

Wealth ↔ education: ~0.45

Most numeric variables are weakly correlated — each adds unique information.



Data Engineering & Preprocessing

Cleaning, encoding, and preparing model-ready features

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Data Integrity & Cleaning Decisions

Structural Missingness

age_first_birth: 8,813 missing — all women with 0 children.

partner_education: 14,039 missing — women with no partner.

Imputed 0 for age_first_birth for nulliparous women

Created has_given_birth binary flag

Duplicates Resolution

274 rows flagged. All 532 look-alikes have unique caseids.

All 32,156 rows retained for statistical integrity.

Post-Cleaning Validation

```
assert df.isnull().sum().sum() == 0
assert df.shape[0] == 32156
print('Validation passed.')
```

Before & After Summary

Raw: 32,156 rows × 20 cols

Cleaned: 32,156 rows × 21 cols

0 missing values, 0 rows dropped

Every downstream step inherits guaranteed completeness.

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Feature Engineering: 18 New Features

Ordinal Encoding

education, wealth, age_group encoded ordinally for tree models.

Nominal One-Hot Encoding

county, region, religion, marital status → one-hot.

Feature space: 20 raw → 38 engineered → 127 model-ready (post OHE).

Collinearity Resolution

Derived child_loss, surviving_ratio from r=0.98 collinear pair.

Key Engineered Features

has_partner — 1 if partner education exists

education_gap — partner's minus woman's education

child_density — children / household size

arid_county — binary flag for ASAL regions

is_in_union — currently in a union

employed — currently working

age_at_first_birth_gap — years since first birth

All features motivated by EDA findings.

Modeling & Evaluation

Five models benchmarked: logistic regression to self-attention transformers

Model Comparison & Selection

Train-Test Split & Resampling

Stratified 80/20 split. SMOTE on training data only, upsampling minority classes to 14,955 each.

Selection Verdict

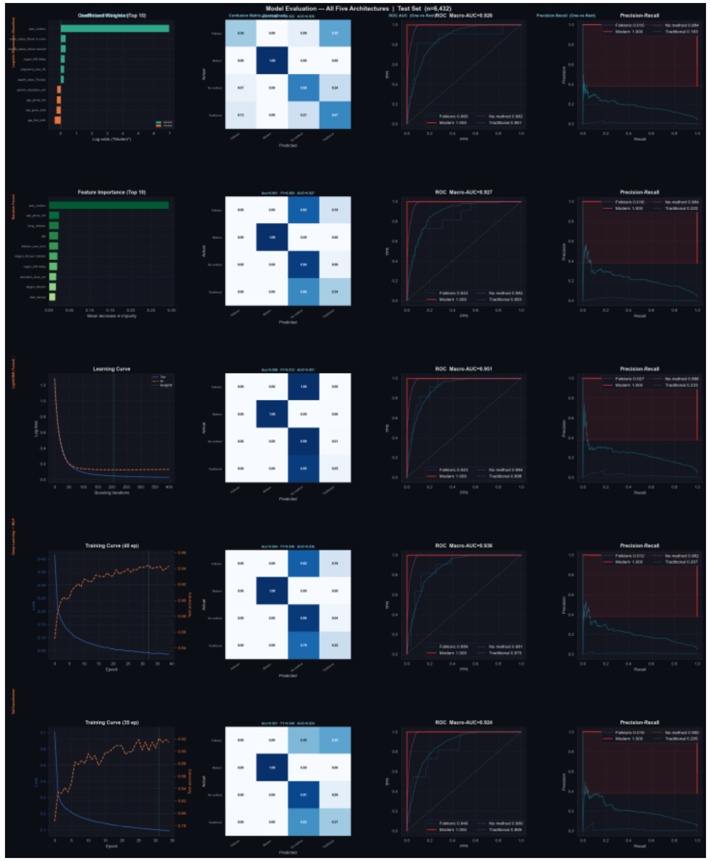
LightGBM selected for deployment: best accuracy (95.8%) and Macro-AUC (0.951).

Notable Finding

TabTransformer achieved highest Traditional-class recall (37%), capturing region×religion interactions.

Model	Accuracy	Macro F1	Macro AUC
LightGBM (Tuned)	**95.8%**	**0.513**	**0.951**
Random Forest	94.1%	0.565	0.927
Deep MLP	94.4%	0.548	0.936
TabTransformer	92.1%	0.549	0.924
Logistic Reg. (Base)	80.8%	0.522	0.926

Side-by-Side Model Evaluation Grid



All 5 models evaluated on identical test set: Confusion Matrix • ROC Curves • PR Curves • Model-specific diagnostics

Final Model: LightGBM (Selected for Deployment)

Performance Summary

Accuracy: 95.8% (best of all 5 models)

Macro-AUC: 0.951 (best of all 5 models)

Modern Recall: 100%

No-method Recall: 99%

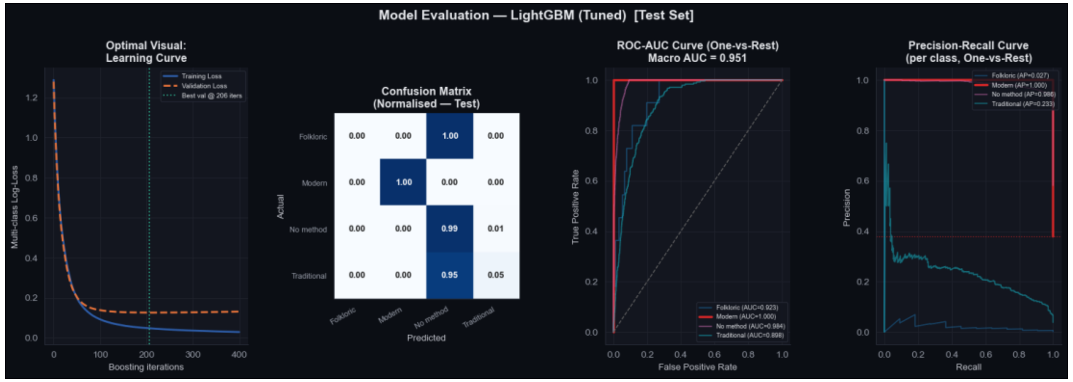
Traditional Recall: 5% (rare-class suppression)

Training Behavior

Validation loss plateaued, confirming no overfitting.

Rare-Class Limitation

Gradient boosting downweights rare-class residuals. Model works best as a binary risk flag.



SHAP Panel A: High vs. Low Feature Groups

What This Shows

For each feature, the red dot shows the average SHAP value for women with HIGH values; blue dot for LOW values.

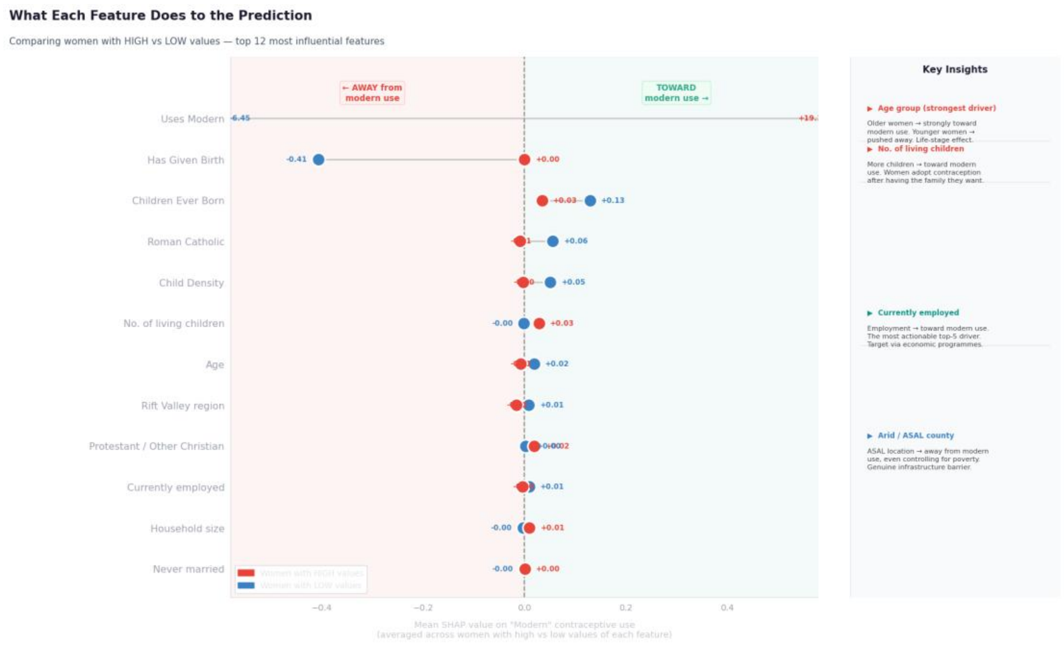
Key Findings

Age: strongest driver — older women pushed toward modern use

Living children: adoption happens after having children

Employment: most policy-actionable lever

is_in_union: strongest enabler of modern use



SHAP Panel B: Global Feature Importance

What This Shows

Average absolute SHAP value for each feature across all classes and samples.

Top Drivers (Ranked)

Age — by far the strongest overall

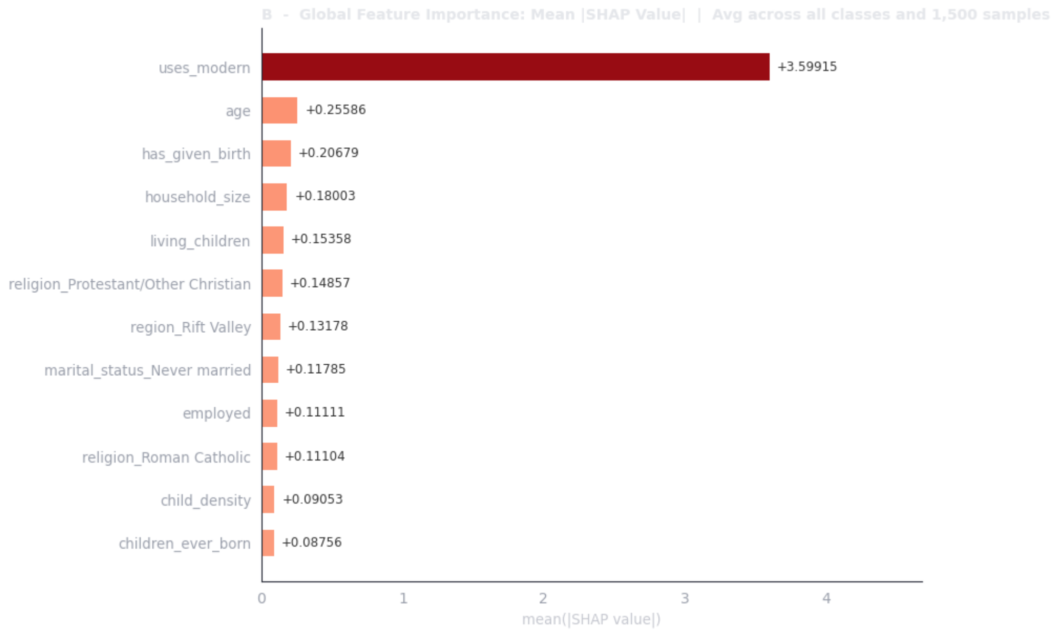
Living children

Region/County (arid_county, region dummies)

Employment status

is_in_union / marital status

Age and living_children are universal discriminators across all four classes.



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SHAP Panels C & D: Two Real Women

Respondent C (“No method”, p=0.9994)

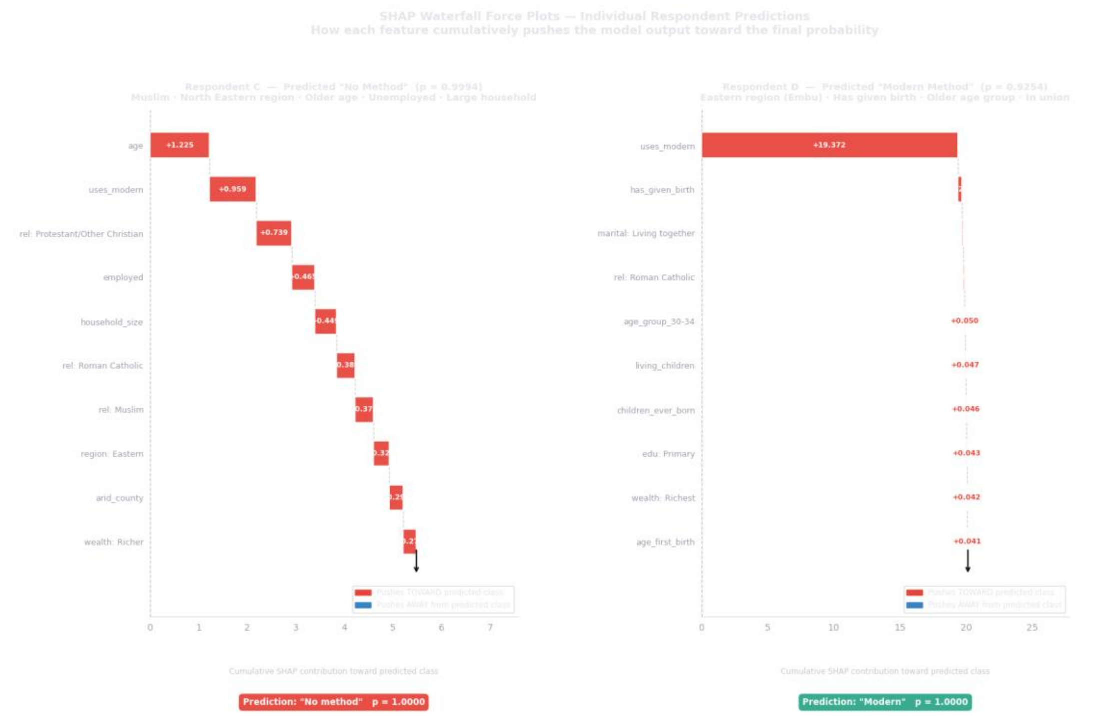
Young, never-married, zero children, no employment. Every feature pushes AWAY from modern use except marital status contributions.

Respondent D (“Modern”, p=1.0000)

Older, married, multiple children, employed. Age and living_children are the dominant positive drivers.

Implication

Individual-level explanations build trust with stakeholders and validate the model’s logic.



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SHAP: Per-Class Breakdown & Direction of Influence

Per-Class Breakdown (Panel E)

Age and living_children are tall bars for every class — universal discriminators.

region and arid_county are disproportionately important for Traditional/Folkloric classes.

Direction Heatmap (Panel F)

is_in_union: deep red for Modern, blue for No-method — strongest enabler

arid_county: red for No-method, blue for Modern — infrastructure barrier

employed: enables Modern use, penalizes No-method

Deployment & Recommendations

From model to actionable policy impact

Deployment: Flask API on Render.com

Interactive Web Application

Deployed on Render.com for stakeholders (Ministry of Health, CHPs, UNFPA).

Serve-Skew Prevention

feature_engineering.py is a shared module imported by both the training notebook and Flask API.

Identical transformations guaranteed — mismatches structurally impossible.

Model Bundle

Saved as contraceptive_model_bundle.joblib (5.62 MB) containing model + preprocessor + label encoder.

Flask API Endpoints

```
GET  /         HTML Form Interface
GET  /health   Server Healthcheck
POST /predict  Single Prediction
```

Live URL

kdhs-contraceptive-api.onrender.com

Auth: X-API-Key header required

Smoke Test Verified

End-to-end: raw JSON → feature_engineering → preprocessing → LightGBM → prediction → API response.

Actionable Policy Recommendations

1. Geographic Prioritization (ASAL Regions)

Mandera (1.2%), Wajir (2.8%), Garissa (4.5%) face systemic infrastructure barriers. Fund mobile clinics and supply chains.

2. Target Exposed Non-Users

4-variable CHP filter: in a union + unemployed + Poorest quintile + arid county. Avoids wasting visits on unexposed young women.

3. Employment Is the Most Actionable Lever

SHAP shows employment pushes women toward modern use. Women's economic empowerment programs should be linked to family planning outreach.

4. Subsidize the Poorest Quintile

Modern use jumps 25% → 41% from Poorest to Poorer, then flattens. Target subsidies at the Poorest exclusively.

5. Model Limitations

Traditional/Folkloric classes too small for reliable individual prediction. Use as binary risk flag.

Thank You!

Questions & Answers

GitHub:
github.com/symekah1999/Predicting-contraceptive-use-among-women

Live App:
kdhs-contraceptive-api.onrender.com